

ALFIDO TECH INTERNSHIP REPORT

Project Title: Model Deployment using FastAPI and Docker

Artificial Intelligence Internship

Abstract

This project demonstrates deployment of a trained image classification model using FastAPI and Docker. The trained model is exposed through REST API endpoints for inference and containerized for easy deployment.

Objectives

1. Deploy a trained AI model using FastAPI.
2. Create REST API endpoints for inference.
3. Containerize the application using Docker.
4. Demonstrate local deployment and testing.

Tools and Technologies

Python 3.x, FastAPI, Uvicorn, PyTorch, Torchvision, Pillow, Docker.

Project Architecture

Client -> FastAPI Endpoint -> Trained Model -> Prediction Response

API Endpoints

GET / : Health check endpoint.

POST /predict : Accepts an image and returns predicted class and confidence score.

Implementation Details

The pretrained ResNet18 model is loaded from model.pth. Uploaded images are preprocessed, passed through the model, and the predicted class is returned as JSON.

Docker Containerization

A Dockerfile packages the application and its dependencies. The container exposes port 8000 and runs the FastAPI server.

Example Request

```
curl -X POST -F "file=@sample.jpg" http://localhost:8000/predict
```

Example Response

```
{ "prediction": "cat", "confidence": 96.45 }
```

Artifacts Produced

app.py
Dockerfile
requirements.txt
README.md
model.pth
screenshots

Folder Structure

```
ImageClassifierAPI/  
├── app.py  
├── Dockerfile  
├── requirements.txt  
├── README.md  
├── model.pth  
└── screenshots/
```

Responsible AI Considerations

No personal data is used. The model is intended for educational purposes. Documentation and transparency are maintained.

Future Enhancements

Deploy on cloud platforms, add authentication, support batch inference, and monitor performance.

Conclusion

The project successfully demonstrates deployment of a machine learning model using FastAPI and Docker, meeting the Alfido Tech internship requirements.